(DBT) ASSIGNMENT-08

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE cnum = (SELECT cnum FROM customers WHERE cname = 'Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE snum IN ( SELECT snum FROM customers WHERE city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE amt > ANY (SELECT amt FROM orders WHERE cnum IN (SELECT cnum FROM customers WHERE
 city = 'London'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3-98839-Himanshu SELECT * FROM customers
    -> WHERE rating >= ANY(
    -> SELECT rating
    -> FROM customers
    -> WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD3-98839-Himanshu SELECT * FROM customers
    -> WHERE rating > ANY (
    -> SELECT rating FROM customers
    -> WHERE city = 'Rome'
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE amt > ANY( SELECT amt FROM orders WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-98839-Himanshu SELECT * FROM orders
    -> WHERE amt < ANY (
    -> SELECT amt
    -> FROM orders
    -> WHERE cnum IN (SELECT cnum FROM customers WHERE city = 'San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3-98839-Himanshu SELECT o.amt, c.cname , c.rating  FROM orders o
    -> INNER JOIN customers c ON c.cnum = o.cnum
    -> WHERE amt > (SELECT AVG(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.00 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
KD3-98839-Himanshu  SELECT * FROM customers
    -> WHERE rating >ALL(SELECT rating FROM customers
    -> WHERE city = 'London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD3-98839-Himanshu SELECT * FROM salespeople s
    -> WHERE snum not in (SELECT snum FROM customers WHERE city = s.city );
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3-98839-Himanshu SELECT sname , snum FROM salespeople s
    -> WHERE (SELECT COUNT(cnum) FROM customers WHERE snum = s.snum )>1;
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3-98839-Himanshu  SELECT s.snum, s.sname, s.city, COUNT(c.cnum) AS customer_count
    -> FROM salespeople s
    -> JOIN customers c ON s.snum = c.snum
    -> GROUP BY s.snum, s.sname, s.city
    -> HAVING COUNT(c.cnum) > 1;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table

```
KD3-98839-Himanshu SELECT snum, SUM(amt) AS total_sales_amount
    -> FROM orders
    -> GROUP BY snum
    -> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
+------+--------------------+
| snum | total_sales_amount |
+------+--------------------+
| 1001 |           15382.07 |
+------+--------------------+
1 row in set (0.00 sec)
```